WEEK 3 TASK: REAL-WORLD MINI PROJECT

Design and Assembly of a Pipe Vice Using SOLIDWORKS

Internship: Mechanical Design Intern

Organization: YuvaIntern

Software Used: SOLIDWORKS

Project Type: Mechanical Design and Assembly

1. Introduction

A pipe vice is a mechanical device used to hold pipes securely during operations such as cutting, threading, and maintenance. It provides stability and reduces unwanted movement while working on a pipe.

In this project, I created a 3D model and assembly of a pipe vice using SOLIDWORKS. The project helped me understand the design of individual mechanical components, their assembly, and the working principle of a screw-operated clamping mechanism.

2. Objective

- To develop a practical mechanical design project using SOLIDWORKS.
- To create and assemble different components of a pipe vice.
- To understand the working principle of a screw-operated clamping mechanism.
- To practise 3D part modelling, component positioning, and assembly mates.
- To understand how mechanical components work together in a functional assembly.

3. Problem Statement

Pipes need to be held firmly during cutting, threading, and other workshop operations. If a pipe is not secured properly, it may move during the operation and affect the accuracy and safety of the work.

The objective of this project is to design a pipe vice assembly that demonstrates how a mechanical clamping mechanism can hold a pipe securely.

4. Components of the Pipe Vice

The main components included in the model are:

1. Fixed Jaw: Provides a stationary clamping surface to support and hold the pipe.
2. Movable Jaw: Moves to clamp or release the pipe during operation.
3. Mounting Shaft: Supports or guides the relevant moving component within the assembly.
4. Mounting Pin: Connects and retains components while allowing the intended movement, depending on its position.
5. Spindle Screw: Converts rotational motion into linear movement to operate the clamping mechanism.
6. Connecting Wedge: Transfers movement or force within the clamping mechanism.
7. Knob: Provides a gripping point for manually rotating the spindle screw.
8. Mounting Pin: Secures another joint or component in the assembly, depending on its location

5. Design and Modelling Process

The pipe vice was modelled and assembled in SOLIDWORKS using the following approach:

1. Created individual 3D components using suitable sketches and dimensions.

2. Used features such as Extruded Boss/Base, Extruded Cut, Revolved Boss/Base, and Fillet wherever required.

3. Designed the main base and supporting components to provide structural support.

4. Modelled the threaded screw and handle to represent the manual clamping mechanism.

5. Inserted the components into an assembly and positioned them using appropriate mates.

6. Checked the overall arrangement and appearance of the completed assembly.

6. Working Principle

The pipe vice works on the principle of a screw-operated clamping mechanism. When the handle is rotated, the screw moves linearly through its threaded arrangement. This movement applies clamping force through the jaw, helping secure the pipe against the supporting surface.

The base provides support, while the frame maintains the relative position of the main components. The screw mechanism allows the clamping position to be adjusted according to the pipe size within the design's intended range.

7. Applications

Pipe vices are commonly used in:

- Mechanical workshops
- Plumbing and pipe-fitting work
- Pipe cutting and threading operations
- Maintenance and fabrication activities
- Industrial repair work

8. Challenges Faced

During the design and assembly process, attention was required to position the components correctly and maintain suitable alignment between the screw, handle, supporting frame, and clamping jaw.

The project also required understanding how individual parts fit together and how the screw mechanism produces clamping movement.

9. Learning Outcomes

Through this project, I practised:

- 3D mechanical part modelling in SOLIDWORKS.
- Creating and assembling multiple components.
- Applying assembly mates and checking component alignment.
- Understanding screw mechanisms and clamping principles.
- Visualising a practical workshop tool through CAD modelling.
- Presenting a mechanical design project through technical documentation.

10. Conclusion

The pipe vice project provided practical experience in mechanical design and assembly using SOLIDWORKS. It helped me understand the relationship between individual components and the working principle of a screw-operated clamping mechanism.

This mini project strengthened my understanding of 3D modelling, assembly design, component positioning, and the application of CAD tools to a real-world mechanical device.

11. Project Images

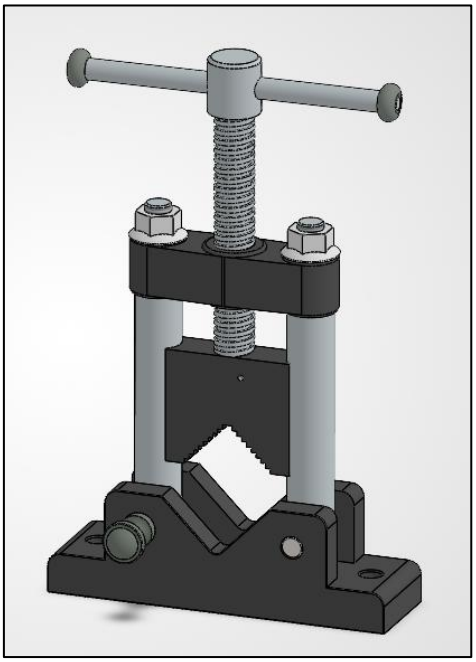


Figure 1: Isometric view of the completed Pipe Vice Assembly.

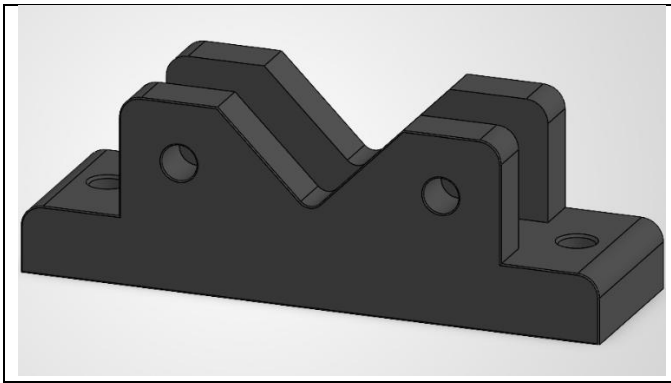


Figure 2: Fixed Jaw

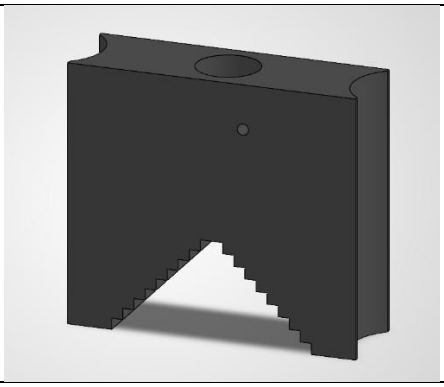


Figure 3: Movable Jaw

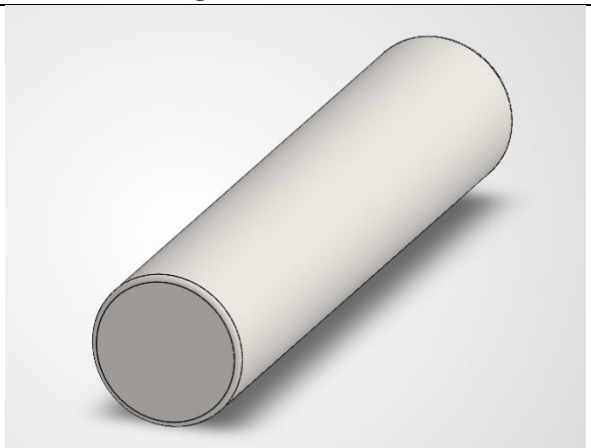


Figure 4: Mounting Pin

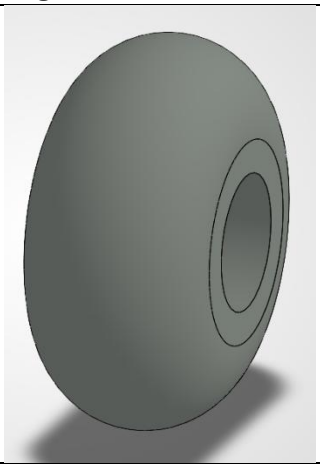


Figure 5: Knob

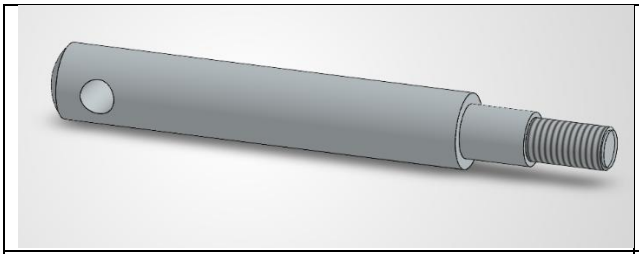
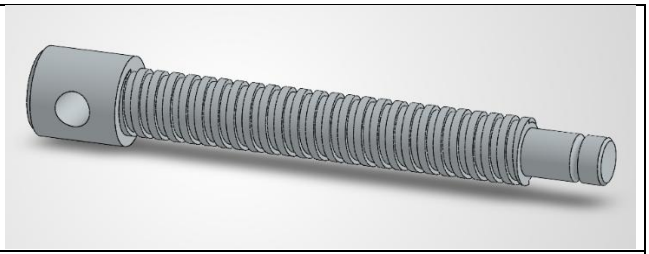
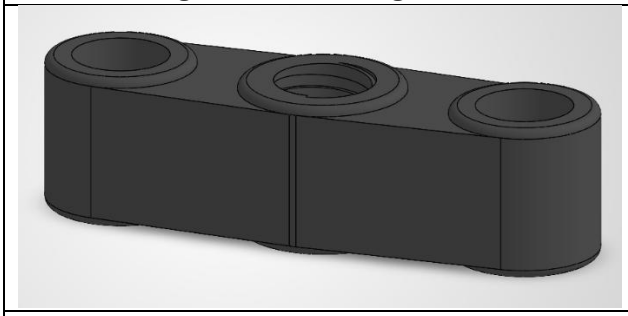
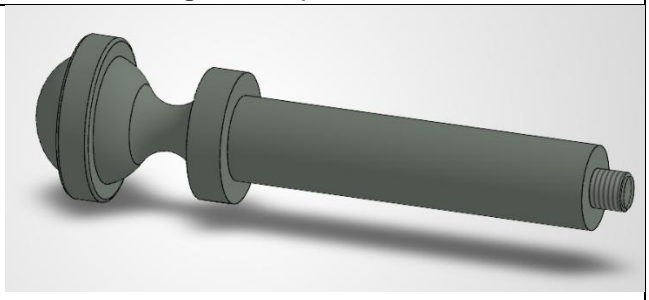
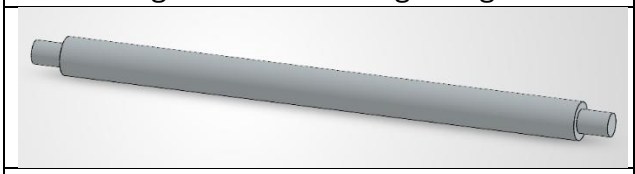
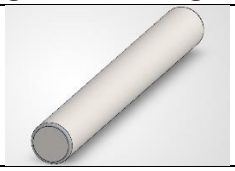
	
Figure 6: Mounting Shaft	Figure 7: Spindle Screw
	
Figure 8: Connecting Wedge	Figure 9: Moving Pin
	
Figure 10: Turning Shaft	Figure 11: 3mm Pin

Table 1: Individual components of the Pipe Vice..